

While Loop

The while loop repeats a block of code as long as the specified condition is true.

Syntax

```
while (condition) {  
    // code block to be executed  
}
```

Example

```
int i = 0;  
  
while (i < 5) {  
    System.out.println(i);  
    i++;  
}
```

the code in the loop will run again and again, as long as a variable (i) is less than 5

Do/While Loop

This loop will execute the code block once, before checking if the condition is true. Then it will repeat the loop as long as the condition is true.

Syntax

```
do {  
    // code block to be executed  
}
```

while (condition);

Example

The example below uses a do/while loop. The loop will always be executed at least once, even if the condition is false, because the code block is executed before the condition is tested

int i = 0;

do {

System.out.println(i);

i++;

}

while (i < 5);

Enhanced for loop

The enhanced for loop (also known as the for-each loop) in Java is a simplified, more readable way to iterate through every element of an array or a collection without using an explicit loop counter or index variable. It was introduced in Java 5 to eliminate manual index tracking and prevent off-by-one errors.

Syntax

java

for (dataType element : arrayOrCollection) {

// Code to execute for each element

```
}
```

Example

```
int[] numbers = {10, 20, 30, 40};
```

```
for (int num : numbers) {
```

```
    System.out.println(num);
```

```
}
```